

SACHIN PATEL

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PROFILE SUMMARY

Operations Research Scientist with 1.5+ years designing and deploying production MILP and Vehicle Routing Problem (VRP) optimization models for large-scale logistics operations. Experienced across the full optimization stack — Pyomo, Gurobi, CBC, GLPK, OR-Tools, and PuLP — with hands-on work in scheduling, network optimization, and constraint-based decision systems. Combines strong mathematical modeling with practical software engineering: SQL/MongoDB ETL pipelines, backend APIs, and applied machine learning for forecasting. Looking to bring this optimization and scheduling background to an Applied Scientist / Optimization Engineer role.

KEY SKILLS

Optimization & Mathematical Programming: MILP, Linear & Integer Programming, Vehicle Routing Problem (VRP), Network Optimization, Constraint Programming, Scheduling, Workforce Planning, Pyomo, Gurobi, CBC, GLPK, OR-Tools, PuLP

Programming & Data Engineering: Python, SQL, MongoDB, Pandas, NumPy, Git, Airflow, REST APIs

AI & Data Science: Scikit-learn, XGBoost, Statistical Modeling, Forecasting, Large Language Models (LLMs), Prompt Engineering

Supply Chain & Analytics: Route Planning, GPS Analytics, Demand Forecasting, Data Validation, Decision Support Systems, Streamlit

EXPERIENCE

Operations Research Scientist — Caliper Business Solutions

Jul 2024 – Jan 2025

- Designed and deployed MILP and VRP optimization models in Pyomo (Gurobi in production, CBC/GLPK for prototyping) for large-scale logistics planning, cutting fleet operating costs and improving vehicle utilization by 10% across 150+ daily routes
- Built backend Python APIs for GPS-based route analytics, including data validation and frequent-path detection across 50K+ GPS records, reducing route-analysis turnaround time by 30%
- Developed automated SQL and MongoDB ETL pipelines to support daily optimization workflows without data delays
- Applied Scikit-learn models to improve supply-chain forecasting accuracy, and deployed modular PuLP/OR-Tools components to production under Git version control

Operations Research Intern — Caliper Business Solutions

May 2023 – Jun 2024

- Built customized VRP models in Pyomo integrating commercial (Gurobi) and open-source (CBC/GLPK) solvers, cutting route-planning time from 6 hours to 2 hours and enabling daily schedule updates
- Automated data collection, cleaning, and preprocessing pipelines in Python; validated routing strategies later scaled into production, contributing to a ~10% reduction in vehicle usage and operating costs

PROJECTS

Hospital Resource Allocation Optimizer | [Live Demo](#)

- Built a Mixed Integer Programming model in Pyomo for hospital workforce scheduling and bed allocation, balancing staffing requirements, labor constraints, and patient demand

Dynamic Pricing Engine | [Live Demo](#)

- Built a pricing optimization engine combining demand forecasting, price-elasticity modeling, and mathematical optimization, delivering simulated profit improvements of up to 22% across tested scenarios

EV Charging Grid Orchestration | [Live Demo](#)

- Designed an EV charging infrastructure planning tool applying demand clustering (DBSCAN) and Set Cover optimization across 500+ simulated demand points, achieving 90%+ geographic coverage while minimizing station count

EDUCATION

M.Sc., Industrial Engineering & Operations Research — Indian Institute of Technology Bombay

Jul 2022 – May 2024

CGPA: 8.6/10

B.Sc., Operational Research — Deen Dayal Upadhyaya College, University of Delhi

Jul 2019 – May 2022

CGPA: 8.8/10

ACHIEVEMENTS

- AIR 60, IIT JAM Mathematics (2022)
- Elite Silver, NPTEL Data Science for Engineers
- JNV Scholar, selected via Jawahar Navodaya Vidyalaya entrance exam